# Study guide: device-independent bit commitment from CHSH

Reconstruction workbook for Aharon, Massar, Pironio, Silman (NJP 18, 025014, 2016)

## Contents

# 0. How to use this guide

## What "build the paper yourself" means

The finished paper is a *report* of a finished argument. This guide is the argument in the order you should *discover* it. If you complete every checkpoint, you will have:

1. The cryptographic definitions and the DI (device-independent) threat model.

2. An honest physical implementation (EPR pairs + four measurement settings per box).
3. The three-phase protocol, including *why* the timing and the private random $n$ are there.
4. A no-signalling proof that $P_{\text{gain}} \le 3/4$, with an explicit saturating strategy.
5. An SDP formulation of Alice's control at a fixed CHSH value $I$, plus an explicit two-qubit strategy that saturates it at $P_{\text{cont}} = \cos^2(\theta/2)$.
6. A martingale / Azuma–Hoeffding finite-$N$ bound that recovers the same number as $N \to \infty$.
7. The PR-box (post-quantum) protocol and the two "free reveal time" variants.

You can then write the paper from your notes using 11-write-the-paper.md.

### How to see mathematical signs

The chapters use `$inline$` and `$$display$$` math (GitHub / KaTeX / MathJax). They should render in GitHub and in Cursor preview. If you want a fully typeset copy:

- HTML: build/study-guide.html
- PDF with Unicode math font: build/study-guide.pdf
- Symbol card: unicode-math.md (π, θ, ≤, ⊕, ⊗, √, ∞, …)

Rebuild after editing:

```bash
bash scripts/build_study_guide.sh
```

### Suggested working method

- Keep a notebook with numbered lemmas of *your* making. Do not copy the paper's numbering until the write-up stage.
- After each chapter, close the guide and try to restate the chapter's claim in one paragraph without looking.
- Only then open solutions.md for that chapter.
- Use the reconstruction-checklist.md as a "paper completeness" test before you start writing.

### Recommended order (do not skip)

| Order | Chapter | Paper analogue | Output of the chapter |
|---|---|---|---|
| 1 | 01 Prerequisites | implicit | CHSH, Tsirelson, PR boxes, POVMs, martingales |
| 2 | 02 Research question | §1 | one-sentence thesis of the paper |
| 3 | 03 Bit commitment | §2.1 | $P_{\text{cont}}$, $P_{\text{gain}}$, balance |

| Order | Chapter | Paper analogue | Output of the chapter |
|---|---|---|---|
| 4 | 04 Device-independence | §2.2 | the five assumptions and what “sending a box” means |
| 5 | 05 Honest resources | §3 intro | measurement table and both families of correlations |
| 6 | 06 Protocol | §3 | the protocol, timing constraints, abort conditions |
| 7 | 07 Alice’s security | §4 | $P_{\text{gain}} \le 3/4$ |
| 8 | 08 Bob’s security, $N = \infty$ | §5.1–5.2 | $C(I)$ and the optimal cheating strategy |
| 9 | 09 Bob’s security, finite $N$ | §5.3, App. D | the bound (19) |
| 10 | 10 Appendices | Apps. A–C | PR-box protocol; free reveal time |
| 11 | 11 Write the paper | whole | section-by-section writing plan |

### What you should *not* do

- Do not start by optimizing Alice’s four-outcome POVM. First understand why a single two-outcome measurement already saturates the bound.
- Do not treat the random index $n$ as a technicality. If Bob does not hide $n$, Alice cheats perfectly.
- Do not confuse “CHSH testing” with “the commit/reveal measurements”. The whole security idea is that the *kept* box cannot tell them apart.

### Notation you should adopt from the start

Fix this now; the paper’s later sections become unreadable if you improvise.

- Two boxes, labelled $i \in \{0, 1\}$.
- Four inputs $s \in \{0, 1, 2, 3\}$, two outputs $r \in \{0, 1\}$.
- $k$-th use of box $i$: random variables $S^i_k$, $R^i_k$; realizations $s^i_k$, $r^i_k$.
- $W_k = \{S^0_k, S^1_k, R^0_k, R^1_k\}$, history $\mathbf{W}_k = \{W_1, \dots, W_k\}$.
- $\sigma_\theta = \cos\theta\,\sigma_z + \sin\theta\,\sigma_x$.
- $|0\rangle, |1\rangle$ are the $\pm 1$ eigenstates of $\sigma_z$.
- Alice’s committed bit $b$; token bit $q$; one-time-pad bit $a$.
- Coin $c \in \{0, 1\}$ chooses which box Alice receives; $\bar{c} = 1 - c$ is the box Bob keeps.

# 1. Prerequisites

Do not open the paper until you can do the exercises in this chapter from memory. Everything later is a cryptographic wrapping of these facts.

## 1.1 Quantum measurements and no-signalling

A two-outcome measurement with setting $s$ on system $i$ is a POVM $\{\Pi^i_{r|s}\}_{r=0,1}$ with $\Pi^i_{r|s} \geq 0$ and $\sum_r \Pi^i_{r|s} = \mathbb{1}$. For a bipartite state $\rho$ and product measurements,

$$P(r^0, r^1 \mid s^0, s^1) = \mathrm{Tr}(\rho\, \Pi^0_{r^0|s^0} \otimes \Pi^1_{r^1|s^1}).$$

**Exercise 1.1.** Prove the no-signalling identities

$$\sum_{r^1} P(r^0, r^1 \mid s^0, s^1) = P(r^0 \mid s^0)$$

independent of $s^1$, and the symmetric one for Bob. This is the *only* constraint used in §4 of the paper (Alice's security). Quantum theory is *not* used there.

**Exercise 1.2.** Why does "the boxes may have clocks, gyroscopes, and memory" not violate the formula above, provided that when two boxes are measured they do not communicate? Write one sentence that will become part of your §2.2.

## 1.2 The CHSH inequality

For binary inputs/outputs $x, y, a, b \in \{0, 1\}$, the CHSH correlator is

$$I = \sum_{a,b,x,y=0,1} (-1)^{a\oplus b\oplus xy}\, P(a, b \mid x, y).$$

Equivalently, if $A_x, B_y$ are $\pm 1$-valued observables ($A_x = 1 - 2a$, etc.),

$$I = \langle A_0 B_0\rangle + \langle A_0 B_1\rangle + \langle A_1 B_0\rangle - \langle A_1 B_1\rangle.$$

Bounds you must know:

| Theory | Bound on $I$ | Name |
|---|---|---|
| Local hidden variables | $\lvert I\rvert \leq 2$ | CHSH / Bell |
| Quantum mechanics | $\lvert I\rvert \leq 2\sqrt{2}$ | Tsirelson |
| No-signalling (no QM) | $\lvert I\rvert \leq 4$ | algebraic / PR-box |

**Exercise 1.3.** Derive the local bound $I \leq 2$ from determinism plus no-signalling (or from the usual CHSH argument).

**Exercise 1.4.** For $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and observables in the $zx$-plane, prove

$$\langle\phi^+|\sigma_\alpha \otimes \sigma_\beta|\phi^+\rangle = \cos(\alpha - \beta).$$

Then recover Tsirelson's bound by placing Alice at $0, \pi/2$ and Bob at $\pi/4, -\pi/4$ (or an equivalent pair). Write down the four angles you used; you will reuse this geometry in §5.2.

**Exercise 1.5.** A *CHSH indicator* for a single round with uniformly random inputs is

$$I(W) = 4 \sum_{a,b,x,y} (-1)^{a\oplus b\oplus xy}\, \delta_{A,a}\delta_{B,b}\delta_{X,x}\delta_{Y,y}.$$

Show that $\mathbb{E}[I(W)]$ equals the CHSH expression above. Why the prefactor 4? (This is footnote 3 of the paper.)

### 1.3 Pseudo-telepathy versus statistics

A nonlocal game is *pseudo-telepathic* if there is a quantum (or no-signalling) strategy that wins with probability 1, while every classical strategy wins with probability $< 1$.

**Exercise 1.6.** The GHZ game is pseudo-telepathic. CHSH is not: even the Tsirelson strategy wins with probability $\cos^2(\pi/8) \simeq 0.85$, not 1. Why does that matter for *distrustful* cryptography, where Bob must test nonlocality *and* check Alice's revealed bit, and the two parties will not cooperate?

Write a four-sentence answer. This *is* the motivation of the paper. Compare with 02-the-research-question.md after you have written it.

### 1.4 PR boxes

A Popescu–Rohrlich box is the no-signalling resource

$$r^0 \oplus r^1 = s^0 \cdot s^1, \qquad r^i, s^i \in \{0, 1\}$$

(up to local relabelling). It achieves $I = 4$.

**Exercise 1.7.** Verify no-signalling for the PR box, and that the CHSH game is won with certainty. This is why Appendix A can drop sequential testing: verification of the box and of the commitment can be the *same* measurement, as in the GHZ protocol.

### 1.5 EPR correlations used as a "commitment token"

The honest protocol needs *two* kinds of correlations from the same pair of boxes:

1. CHSH violation on input pair $\{0, 1\} \times \{0, 1\}$.
2. Perfect equality $r^0 = r^1$ on the "shifted" pairs $(s^0, s^1) = (i, i + 2 \bmod 4)$ for $i = 0, 1, 2, 3$.

**Exercise 1.8 (do this before reading §3).** Invent four angles for box 0 and four for box 1, all of the form $\sigma_\theta$, such that:

- inputs $(0, 1)$ on both boxes give the Tsirelson CHSH strategy;
- for each $i$, the observable of box 0 at input $i$ equals the observable of box 1 at input $i + 2 \bmod 4$.

There is a canonical solution (the paper's). Finding it yourself makes the commit/reveal rule obvious: Alice's commit input is $b + 2$, Bob's check input is $b$, and they are measuring the *same* observable.

### 1.6 Semidefinite relaxations of quantum correlations

Alice's control at fixed $I$ is a polynomial optimization problem over unknown Hilbert-space operators. The NPA hierarchy (Navascués–Pironio–Acín) produces a sequence of SDPs whose optima decrease to the true quantum value.

**Exercise 1.9.** Read enough of NPA (PRL **98**, 010401 (2007) or NJP **10**, 073013 (2008)) to answer:

- What is a *behaviour* $P(ab|xy)$?
- What is a *moment matrix* at level 1 and at level 2?
- Why does a feasible moment matrix *not* prove that a quantum realization exists, while infeasibility *does* prove that none exists?

You do **not** need to re-derive NPA. You need to know that the paper solves the *level-2* relaxation of (7), and then exhibits an explicit strategy whose $P_{\text{cont}}$ matches the SDP number to $10^{-8}$. Matching means the relaxation has already converged.

### 1.7 Martingales and Azuma–Hoeffding

Let $\{\mathcal{F}_k\}$ be a filtration (think: the history $\mathbf{W}_k$). A process $Z_k$ is a martingale if $\mathbb{E}[Z_{k+1} \mid \mathcal{F}_k] = Z_k$.

**Azuma–Hoeffding.** If $|Z_{k+1} - Z_k| \le D$ almost surely, then for $\varepsilon > 0$,

$$P(Z_k - Z_0 \ge k\varepsilon) \le \exp(-k\varepsilon^2/(2D^2)).$$

**Exercise 1.10.** Define

$$\Delta_k = \bar{I}_k - \frac{1}{k}\sum_{n=1}^{k} \mathbb{E}[I(W_n) \mid \mathbf{W}_{n-1}], \qquad Z_k = k\Delta_k.$$

Prove that $Z_k$ is a martingale. Bound $|Z_{k+1} - Z_k|$ using only $|I(w)| \le 4$ and $|\mathbb{E}[I]| \le 2\sqrt{2}$. You should get $D = 4 + 2\sqrt{2}$. This is Appendix D.

### 1.8 Bit-commitment folklore (one hour of reading)

Read, or at least know the statements of:

- Lo–Chau, PRL **78**, 3410 (1997) and Mayers, PRL **78**, 3414 (1997): no *perfect* quantum bit commitment.
- Spekkens–Rudolph, PRA **65**, 012310 (2001): imperfect BC is possible.
- Chailloux–Kerenidis, FOCS 2011: in any *balanced* quantum BC, $P_{\text{cont}} = P_{\text{gain}} \gtrsim 0.739$.
- Silman *et al.*, PRL **106**, 220501 (2011): DI bit commitment from GHZ, with $P_{\text{cont}} = \cos^2(\pi/8)$, $P_{\text{gain}} = 3/4$.
- Kent, PRL **83**, 1447 (1999): *relativistic* BC is perfect, but needs two remote labs for at least one party.

**Exercise 1.11.** In one paragraph, distinguish device-independent BC from relativistic BC. The paper's point is: one lab per party, no spacelike-separation requirement, shielding instead.

### Checkpoint

You are ready for chapter 2 if you can, without notes:

1. Write the CHSH expression and the three bounds 2, $2\sqrt{2}$, 4.
2. Explain why GHZ is pseudo-telepathic and CHSH is not.
3. State $P_{\text{cont}}$ and $P_{\text{gain}}$ in words.
4. Recite Azuma–Hoeffding.

## 2. The research question

This chapter is the paper's §1. Write your own introduction *after* you can answer the questions below in complete sentences.

### 2.1 What problem is open?

By 2015, device-independent (DI) cryptography based on *CHSH testing of bipartite boxes* was known for:

- DIQKD,
- randomness expansion / amplification,
- self-testing,
- entanglement estimation / quantification.

Two bipartite *distrustful* tasks—bit commitment and coin flipping—had DI protocols, but only from **tripartite GHZ** correlations (Silman *et al.* 2011; Aharon *et al.* 2014). No CHSH-based DI protocol was known.

**Exercise 2.1.** Distrustful cryptography means the parties do not trust each other and may have conflicting goals. Why is "certify nonlocality" harder here than in QKD? Write the obstruction in one sentence. (Hint: in QKD, Alice and Bob cooperate against Eve. Here Alice may be Eve.)

**Exercise 2.2.** In the GHZ protocol, Bob uses *the same measurements* both to test nonlocality and to check the revealed bit. Why is that possible? Why does the same trick fail for CHSH?

## 2.2 The thesis you must prove

Formulate the paper as a single claim:

> There exists a DI bit-commitment protocol whose only Bell test is sequential CHSH on a pair of boxes, which in the limit of infinitely many tests achieves the same cheating probabilities as the GHZ protocol of Silman *et al.*, even when the devices have long-term quantum memory.

Add the two caveats the paper is honest about:

1. The reveal time is *fixed* by the protocol (so strictly speaking this is a BC-like primitive, not BC with Alice-chosen reveal time).
2. For noisy devices the honest abort probability in the reveal phase is nonzero—true of any practical BC, DI or not.

**Exercise 2.3.** Why is a fixed reveal time still useful? Give the coin-flipping reduction in two sentences.

## 2.3 The one idea that makes CHSH work

GHZ gives *certainty*. CHSH gives *statistics*. So Bob cannot test the box in one shot and simultaneously get a deterministic check of Alice's bit.

The replacement idea:

> Hide from the devices *which round* is the commit/reveal round.

Concretely:

- Bob chooses a private uniform $n \in \{1, \dots, N\}$.
- He CHSH-tests for $n$ rounds at publicly scheduled times $t_1, \dots, t_n$.
- He sends Alice one box, they commit/reveal, and he measures the kept box at the *next* scheduled time $t_{n+1}$.

Unless $n = N$, the kept box sees "another measurement at the usual time", not "the verification of a commitment".

**Exercise 2.4.** Suppose $n$ is *public*, or Alice's boxes contain a counter that is allowed to change behaviour on use number $N + 1$. Describe a perfect cheating strategy for Alice. This is why the $1/N$ term appears in the finite-$N$ bound.

**Exercise 2.5.** Suppose the time gaps $t_{i+1} - t_i$ are shorter than (send box to Alice) + (Alice measures) + (Alice's classical message returns) + (Bob measures). What can Alice's remaining box infer, and how does she cheat?

## 2.4 What "same security as GHZ" means numerically

Silman *et al.* 2011:

$$P_{\text{cont}} = \cos^2(\pi/8) \simeq 0.8536, \qquad P_{\text{gain}} = 3/4.$$

You will re-obtain both numbers. Note that this protocol is *not* balanced ($0.85 \neq 0.75$), and is therefore *not* trying to meet the Chailloux–Kerenidis 0.739 bound, which applies only to balanced BC.

**Exercise 2.6.** Is $P_{\text{cont}} = \cos^2(\pi/8)$ an accident of GHZ geometry, or a Tsirelson-type angle? After chapter 8 you should see it is the correlation probability of two equatorial observables $\pi/4$ apart on $|\phi^+\rangle$.

### 2.5 Post-quantum subplot

The paper also asks: if honest *and* dishonest parties are limited only by no-signalling, can one do better? Appendix A uses a PR box, restores pseudo-telepathy, and gets a *balanced* protocol with $P_{\text{cont}} = P_{\text{gain}} = 3/4$, which is better for Alice's security *and* for balance, though still not the CK lower bound (that bound is quantum-specific).

**Exercise 2.7.** Why might one have expected *worse* security in a PR-box world? Why might one have expected *better*? The paper's answer is "on balance, better". You should be able to say that in the introduction in three sentences.

### 2.6 Outline you should impose on the paper

When you write §1, end with this roadmap (the paper's):

- §2 definitions (BC + DI assumptions)
- §3 protocol
- §4 Alice's security ($P_{\text{gain}}$)
- §5 Bob's security ($P_{\text{cont}}$, asymptotic then finite)
- §6 summary
- App. A PR-box
- App. B free reveal time, worse $P_{\text{cont}}$
- App. C large-office, free reveal time, same $P_{\text{cont}}$
- App. D Azuma–Hoeffding lemma

### Checkpoint

Write, in at most 250 words, an abstract. Then compare with the published abstract. Yours should mention: GHZ-only previous work, CHSH protocol, memory, same numbers, fixed reveal time, post-quantum appendix.

## 3. Bit commitment

Paper analogue: §2.1. This chapter is short on purpose. Get the definitions exact; every later formula is an instance of them.

## 3.1 The primitive

Two remote parties. Alice has a bit $b \in \{0, 1\}$. A protocol has two phases:

1. **Commit.** Alice sends Bob a token (here: a classical bit $q$). After this, she should be unable to change $b$.
2. **Reveal.** Alice sends information that lets Bob output a bit $\hat{b}$. He should accept $\hat{b} = b$ if both were honest, and he should have learned (almost) nothing about $b$ before this phase.

Classical BC with unbounded computation is impossible (the commit token is either independent of $b$, so Alice opens both ways, or it determines $b$, so Bob reads it). Quantum BC cannot be *perfect* (Lo–Chau, Mayers) but can be *imperfect* (Spekkens–Rudolph).

## 3.2 Cheating probabilities

Assume Alice is equally likely to want to open 0 or 1 after commit (worst-case / uniform).

- $p_0$: probability that dishonest Alice, after commit, successfully opens 0 (Bob does not abort and outputs 0).
- $p_1$: likewise for 1.
- **Alice's control**

$$P_{\text{cont}} = \frac{1}{2}(p_0 + p_1).$$

- **Bob's information gain** $P_{\text{gain}}$: probability that dishonest Bob, using the commit token and his systems, correctly guesses $b$ *before* reveal.

Perfect BC: $P_{\text{cont}} = P_{\text{gain}} = 1/2$.

**Balanced** BC: $P_{\text{cont}} = P_{\text{gain}}$. Chailloux–Kerenidis: any balanced *quantum* BC satisfies $P_{\text{cont}} = P_{\text{gain}} \gtrsim 0.739$, and this is saturable.

**Exercise 3.1.** This paper's asymptotic numbers are 0.8536 and 0.75. Is the protocol balanced? Should you mention the 0.739 bound in the introduction, and if so, how (as a comparison, not as a target)?

**Exercise 3.2.** $P_{\text{cont}}$ is *not* "Alice guesses Bob's measurement". It is "Alice can *choose* which bit to open after commit, and be accepted". Keep this distinction when you design her POVM in chapter 8: she may use different openings for $b = 0$ and $b = 1$, and $P_{\text{cont}}$ averages them.

## 3.3 Completeness versus soundness (honest abort)

Two different abort events appear in §3:

1. Bob aborts in the *random-selection / CHSH-test* phase because $\bar{I}_n < I_{\text{th}}$. This can happen even with ideal boxes (statistical fluctuation) or because Alice is cheating. Bob cannot tell which. As $N \to \infty$ with a gap between $I_{\text{th}}$ and the honest $I$, this probability vanishes.

2. Bob aborts in the *reveal* phase because the token $q$ is inconsistent or the two boxes disagree. With ideal honest boxes this does not happen, once the CHSH test passed. With noise, it happens with positive probability even if Alice is honest.

**Exercise 3.3.** Write two sentences for the paper explaining that (2) is an implementation issue shared by every practical BC protocol, not a DI artefact.

### 3.4 Coin flipping as a consumer of this primitive

A (weak) coin flip can be built from BC: Alice commits a bit $b_A$; Bob sends a bit $b_B$; Alice reveals; the coin is $b_A \oplus b_B$.

**Exercise 3.4.** Why does a *fixed* reveal time not break this reduction? (Bob can send $b_B$ at a time that is still before the scheduled reveal.)

### Checkpoint

Define $P_{\text{cont}}$ and $P_{\text{gain}}$ without looking. State the Lo–Chau/Mayers no-go in one sentence, and Spekkens–Rudolph/Chailloux–Kerenidis in one sentence each.

## 4. Device-independence

Paper analogue: §2.2. Your security proofs are only as strong as this model. Write it *before* the protocol, exactly as the paper does.

### 4.1 The five assumptions

Reproduce these in your own words, then freeze them.

1. **Boxes.** Each device is a black box: classical input $s$, classical output $r$. An input always produces an output (no losses).
2. **Quantum theory.** Honest *and* dishonest parties are limited by QM. (Appendix A relaxes this to no-signalling only.)
3. **No communication between boxes, on demand.** The parties can prevent the two boxes from talking to each other when they choose. This can be shielding, not spacelike separation.
4. **Private randomness.** Each party has a trusted RNG for the protocol's random choices ($n$, inputs, coins, Alice's pad $a$).
5. **Honest lab leakage.** Nothing leaks out of an honest party's laboratory.

**Exercise 4.1.** Which of these is used in Alice's security (§4) but not in Bob's, and vice versa? Make a table. Example: Alice does *not* test CHSH, so Bob's cheating strategy is an entangled ancilla plus a guess from $q$; assumption 3 between Alice's box and Bob's ancilla is *not* imposed after he sent her the box.

**Exercise 4.2.** Losses are excluded. What would a dishonest party do if a box were allowed to output "no-click"? (You do not need a full loss-tolerant analysis; you need to know why the paper assumes no losses.)

## 4.2 The only constraint on honest boxes

If an honest party holds both boxes and isolates them,

$$P(r^0, r^1 \mid s^0, s^1) = \mathrm{Tr}(\rho\, \Pi^0_{r^0|s^0} \otimes \Pi^1_{r^1|s^1}).$$

A dishonest party chooses $\rho$ and the POVMs. The boxes may depend on time, location, past inputs/outputs, and any stored quantum memory.

**Exercise 4.3.** Write the *memory* generalization: the POVM at step $k$ may depend on $(s_1, r_1, \dots, s_{k-1}, r_{k-1})$. This is the setting of Reichardt–Unger–Vazirani, and it is why sequential CHSH testing plus Azuma is needed rather than i.i.d. Chernoff.

## 4.3 What "sending a box" means

The paper is explicit: nobody posts a laboratory apparatus in the mail.

**Exercise 4.4.** Write the operational meaning: a quantum channel carries a system such that, in an honest run, the state and the POVM elements that used to describe Alice's box now describe Bob's box (or vice versa). Dishonestly, the sender may send anything, including a system entangled with an ancilla they keep.

This is why Bob, when cheating, can keep an ancilla entangled with the box he sends Alice.

## 4.4 Not relativistic bit commitment

Relativistic BC (Kent) gives *perfect* commitment using spacelike separation and (at least) two remote secure labs for one party.

**Exercise 4.5.** List three differences with this protocol:

- number of labs per party;
- whether measurements must be spacelike related;
- whether the no-communication assumption is implemented by relativity or by shielding.

Include this comparison in §2.2 of your write-up. The published NJP text emphasizes shielding; the arXiv text is slightly shorter on this point. Follow NJP.

## 4.5 What DI does *not* protect against

DI does not remove:

- a leaky honest lab,
- a dishonest RNG,
- communication between boxes that the parties failed to prevent,
- the need for a quantum channel.

It *does* remove: known Hilbert-space dimension, promised observables, promised state, calibration of detectors (except losses, which are simply forbidden here).

### Checkpoint

Recite the five assumptions. Explain "sending a box" in one sentence. Explain why shielding is enough for assumption 3 even though many measurements in the protocol are *not* spacelike related.

# 5. Honest resources

Paper analogue: the paragraphs of §3 before the numbered protocol. Design the physics *before* you design the cryptography. If the honest correlations are wrong, commit/reveal will not be consistent with CHSH.

## 5.1 Boxes

One pair of boxes. Each box: inputs $\{0,1,2,3\}$, outputs $\{0,1\}$. Think of inputs $0,1$ as the CHSH pair and inputs $2,3$ as "the same observables, on the other box, shifted by 2".

## 5.2 Target correlations (ideal)

For every use $n$,

$$\sum_{r^0,r^1,s^0,s^1=0,1} (-1)^{r^0\oplus r^1\oplus s^0 s^1}\, P(r^0,r^1 \mid s^0,s^1) = 2\sqrt{2},$$

and, separately,

$$r^0 = r^1 \quad \text{whenever} \quad s^0 = i,\ s^1 = i+2 \pmod 4, \quad i \in \{0,1,2,3\}.$$

**Exercise 5.1.** Why four "equality" pairs, not one? Because Alice will receive a random box $c \in \{0,1\}$ and will commit $b \in \{0,1\}$, using input $s^c = b+2$. Bob must be able to measure the *matching* observable on the other box, which is input $s^{\bar{c}} = b$. Check that $(s^c, s^{\bar{c}}) = (b+2,\ b)$ is always one of the four equality pairs, for both values of $c$.

Work through the table (fill every cell):

| $c$ | $b$ | Alice input $s^c$ | Bob input $s^{\bar{c}}$ | pair $(s^0,s^1)$ | should $r^0 = r^1$? |
|---|---|---|---|---|---|
| 0 | 0 | | | | |
| 0 | 1 | | | | |
| 1 | 0 | | | | |
| 1 | 1 | | | | |

## 5.3 The canonical implementation

Prepare (many copies of) $|\phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Observables $\sigma_\theta = \cos\theta\,\sigma_z + \sin\theta\,\sigma_x$:

| Input | Box 0 | Box 1 |
|---|---|---|
| 0 | $\sigma_x = \sigma_{\pi/2}$ | $\sigma_{\pi/4}$ |
| 1 | $\sigma_z = \sigma_0$ | $\sigma_{3\pi/4}$ |
| 2 | $\sigma_{\pi/4}$ | $\sigma_x$ |
| 3 | $\sigma_{3\pi/4}$ | $\sigma_z$ |

**Exercise 5.2.** Verify:

1. Box 0 on $\{0,1\}$ and box 1 on $\{0,1\}$ is a Tsirelson pair: angles $\pi/2, 0$ versus $\pi/4, 3\pi/4$. Compute $I$ using Exercise 1.4. You want $+2\sqrt{2}$ with the CHSH sign pattern $(-1)^{xy}$. If you get $-2\sqrt{2}$, a local relabelling of an outcome is needed; decide which one and record it. (The paper's output labels are chosen so that (2) holds as written.)
2. Box 0 input $i$ equals box 1 input $i+2 \bmod 4$, as operators. Therefore equality of outcomes on $|\phi^+\rangle$ is certain (same observable, singlet-free EPR).

**Exercise 5.3.** Draw the four axes in the $zx$-plane. This figure is not in the paper, but it will make Fig. 2 (Alice's cheating axes) easy later.

## 5.4 Noise

In a real run, $I < 2\sqrt{2}$ and the equality pairs are only *approximately* correlated. The protocol therefore uses a threshold $I_{\text{th}}$ and will sometimes abort on honest equality tests. The security theorems are stated in terms of the *observed* $\bar{I}_n$ and of a function $C(I)$ giving Alice's control at CHSH value $I$.

**Exercise 5.4.** Decide, for your write-up, whether honest completeness is:

- asymptotic (ideal boxes, $N \to \infty$, $\bar{I}_n \to 2\sqrt{2} > I_{\text{th}}$), or
- noisy (you would need a gap $I_{\text{honest}} - I_{\text{th}}$ and a Chernoff/Azuma bound on honest abort).

The paper discusses this qualitatively in §3 and does not give a completeness theorem with explicit $\varepsilon(N)$. If you want a stronger paper, adding one is a natural extension; it is *not* required to reconstruct the published result.

## Checkpoint

From memory, write the measurement table and the rule $s^c = b+2$, $s^{\bar{c}} = b$. Explain in one sentence why those two inputs measure the same observable.

# 6. The protocol

Paper analogue: §3. Write the protocol only after chapters 4–5. Then justify every random choice and every clock tick.

## 6.1 Parameters

A family of protocols, indexed by:

- integer $N > 1$ (maximum number of CHSH tests);
- public times $t_1 < \cdots < t_{N+1}$;
- public threshold $I_{\text{th}}$.

The gaps $t_{i+1} - t_i$ must be long enough for: Bob → Alice quantum transmission, Alice's measurement, Alice → Bob classical messages (commit *and* reveal), Bob's measurement of the kept box at $t_{i+1}$.

Footnote in the paper: it is enough that measurement $i+1$ occur at some time in $(t_i, t_{i+1}]$, provided the remainder of the interval still fits commit+reveal.

## 6.2 Write the protocol yourself

Using only the ingredients below, write a three-phase protocol. Then compare with the paper.

**Phase 1 — Random selection (Bob).**

- Private uniform $n \in \{1, \dots, N\}$.
- Private uniform CHSH inputs $\mathbf{s}_n^0, \mathbf{s}_n^1 \in \{0,1\}^n$.
- At each $t_i$, $i = 1, \dots, n$, input $s^i$ into box $i$.
- Compute

$$\bar{I}_n(\mathbf{w}_n) = \frac{1}{n} \sum_{k=1}^{n} I(w_k)$$

with the indicator (4) of the paper (prefactor 4; see Exercise 1.5). - Abort if $\bar{I}_n < I_{\text{th}}$. - Else flip a coin $c \in \{0,1\}$, and at time $t^b < t_{n+1}$ send box $c$ to Alice.

**Phase 2 — Commit (Alice).**

- Bit $b$. Input $s_{n+1}^c = b + 2$. Obtain $r_{n+1}^c$.
- Private uniform pad $a \in \{0,1\}$.
- At time $t^c \in (t^b, t_{n+1})$, send $q = r_{n+1}^c \oplus ab$.

**Phase 3 — Reveal (Alice then Bob).**

- At time $t^d \in (t^c, t_{n+1})$, Alice sends $b$ and $r_{n+1}^c$. If Bob does not have them before $t_{n+1}$, abort.
- Consistency of the token: accept only if $q = r^c$ or $q = r^c \oplus b$. (Why *or*? Because $q = r \oplus ab$, so for $b = 0$ one must have $q = r$, while for $b = 1$ both $q = r$ and $q = r \oplus 1$ are possible depending on $a$.)

- At time $t_{n+1} > t^d$, Bob inputs $s^{\bar{c}}_{n+1} = s^c_{n+1} - 2 = b$ and aborts unless $r^{\bar{c}}_{n+1} = r^c_{n+1}$.

**Exercise 6.1.** Prove that if Alice is honest, $q = r^c \oplus ab$ always satisfies the token check, for both values of $b$.

**Exercise 6.2.** Prove that if Alice is honest and the boxes implement the table of chapter 5 perfectly, then after a passing CHSH test Bob's last check always succeeds and he outputs the committed bit. (Ideal completeness, conditioned on not aborting in Phase 1.)

**Exercise 6.3.** Why is $q$ a *one-time-padded* version of Alice's outcome, padded *only when* $b = 1$? Connect this to $P_{\text{gain}} = 3/4$: Bob who treats $q$ as $r^c$ is correct whenever $ab = 0$, i.e. three quarters of the time.

## 6.3 The two design constraints you must argue in the text

### Private random $n$

If Alice knows $n$, she programs: uses $1, \dots, n$ are Tsirelson; use $n+1$ is deterministic and lets her open both bits. Bob's CHSH test then always passes and she has $P_{\text{cont}} = 1$.

If $n$ is hidden, the only use she can safely make deterministic is use $N+1$, which happens only when Bob picked $n = N$, probability $1/N$. That is the origin of the $+1/N$ in (11)–(19).

### Fixed times

The kept box must not see a "special" measurement after a long delay. If Bob measured the kept box only *after* receiving Alice's reveal, the box could switch from CHSH-violating to deterministic.

**Exercise 6.4.** Write Alice's perfect cheat if times are *not* fixed. This paragraph is the reason the protocol is "not strictly BC".

**Exercise 6.5.** Coin flipping still works (Exercise 3.4). For true Alice-chosen reveal time, you will need Appendix B or C.

## 6.4 Abort semantics (write this into §3)

- Phase-1 abort: possible even if everyone is honest (statistics). Bob cannot attribute it.
- Phase-3 abort with ideal honest devices: does not happen if Phase 1 passed.
- Phase-3 abort with noisy honest devices: happens with positive probability.

## 6.5 Protocol flowchart (put a version of this in your notes)

```
sequenceDiagram
    participant B as Bob
    participant A as Alice
```

```
    Note over B: pick n, CHSH inputs privately
    loop i = 1..n at times t_i
        B->>B: measure both boxes, collect I(w_i)
    end
    alt bar I_n < I_th
        B->>B: abort
    else
        B->>B: flip c, send box c at t^b
        B->>A: box c
        A->>A: input b+2, sample a
        A->>B: q = r XOR (a AND b) at t^c
        A->>B: (b, r) at t^d < t_{n+1}
        B->>B: check q consistent with (b,r)
        B->>B: at t_{n+1} input b on kept box, check r_bar = r
    end
```

### Checkpoint

Close the guide. Write the protocol from memory, including every time label and every abort condition. If you miss the pad $ab$, the private $n$, or the measurement at $t_{n+1}$, you are not ready for the proofs.

## 7. Alice's security (bound on Bob's information gain)

Paper analogue: §4. This is the easier proof. It uses **only no-signalling**, not Tsirelson's bound, not CHSH testing. That is why it survives unchanged in Appendix A (PR boxes) and in Appendices B–C.

Convention in this chapter: drop the subscript $n+1$. Alice holds box $c$ and inputs $s^c \in \{2,3\}$. Bob does not hold a "CHSH partner" he can trust; he may have kept an ancilla.

### 7.1 Bob's most general cheat

Alice is honest: she picks $b, a$ uniformly, inputs $s^c = b + 2$, sends $q = r^c \oplus ab$.

Bob prepared the box entangled with an ancilla. After seeing $q$, he measures the ancilla with a two-outcome measurement that may depend on $q$.

Notation:

- $m \in \{0,1\}$: which measurement he performs; he will set $m = q$.
- $g \in \{0,1\}$: his guess for $s^c$ (equivalently for $b = s^c - 2$).

The joint law $P(r^c, g \mid s^c, m)$ may depend on all four arguments because of entanglement.

**Exercise 7.1.** Write $P_{\text{gain}}$ as an average over $a, b, r^c$ of the probability that $g = b$. You should obtain the first line of the paper's (5):

$$P_{\text{gain}} = \max_{\mathcal{S}} \sum_{r^c,b,a} P(r^c \mid s^c = b+2)\, P(g = b \mid r^c, s^c = b+2, m = r^c \oplus ab),$$

and then, using $P(a,b) = 1/4$, the second line with a factor $1/4$ and a sum over $r^c, b$ only.

**Exercise 7.2.** Rewrite each term as a joint probability $P(r^c, g = \cdots \mid s^c, m = \cdots)$. Expand the sum over $r^c, b \in \{0,1\}$. You should find only six types of terms, which the paper groups as

$$\frac{1}{4}\max[2P(0,0\mid 2,0)+2P(1,0\mid 2,1)+P(0,1\mid 3,0)+P(1,1\mid 3,1)+P(0,1\mid 3,1)+P(1,1\mid 3,0)],$$

where $P(r,g \mid s,m)$ is abbreviated $P(r,g \mid s,m)$.

(The published algebra is easy to scramble. Do it on paper with a $2\times 2\times 2$ table of $(b,r,a)$.)

## 7.2 No-signalling and normalization

Outputs of Alice's box cannot depend on Bob's later choice of $m$, and Bob's guess $g$ cannot depend on Alice's input $s^c$ if he does not measure a system that she still holds—wait: they *are* entangled, so $P(r,g \mid s,m)$ *can* depend on both. The NS constraints are the *marginals*:

$$\sum_g P(r,g \mid s,m) = P(r \mid s) \qquad \text{(independent of } m\text{)},$$

$$\sum_r P(r,g \mid s,m) = P(g \mid m) \qquad \text{(independent of } s\text{)}.$$

**Exercise 7.3.** Using NS and positivity, prove

$$P(0,0\mid 2,0) + P(1,0\mid 2,1) \le 1$$

and

$$P(m,0\mid 2,m) + P(0,1\mid 3,m) + P(1,1\mid 3,m) \le 1$$

(the paper's form; equivalently a bound on the remaining four terms). Conclude $P_{\text{gain}} \le 3/4$.

Hint for the first inequality: $P(0,0\mid 2,0) \le P(g=0 \mid m=0)$ and $P(1,0\mid 2,1) \le P(g=0 \mid m=1)$, but those are *different* $m$, so that is not enough. Instead compare both terms to Alice-box marginals, or add them as a combination bounded by a NS marginal of Bob. A clean route is:

- $P(0,0 \mid 2,0) + P(1,0 \mid 2,1) \le P(r=0 \mid s=2) + P(r=1 \mid s=2) = 1$ is false in general because the $g$ values differ… wait, first term has $g=0$, second has $g=0$ as well! Both have $g=0$. First has $r=0, m=0$; second has $r=1, m=1$.

Better route used implicitly in the paper: the two terms $P(0,0 \mid 2,0)$ and $P(1,0 \mid 2,1)$ are probabilities of *disjoint* $(r,m)$ pairs but $m$ is chosen by Bob as a function of $r$ in the real protocol, which is already expanded. For the *max over behaviours*, use:

$$P(0,0 \mid 2,0) \le P(g=0 \mid m=0), \qquad P(1,0 \mid 2,1) \le P(g=0 \mid m=1).$$

Those still do not sum to $\le 1$. The paper's stated inequality is $P(0,0 \mid 2,0) + P(1,0 \mid 2,1) \le 1$. Prove it from NS as follows:

$$P(0,0 \mid 2,0) + P(1,0 \mid 2,1) = P(0,0 \mid 2,0) + P(1,0 \mid 2,0) + P(1,0 \mid 2,1) - P(1,0 \mid 2,0).$$

And $P(0,0 \mid 2,0) + P(1,0 \mid 2,0) = P(g=0 \mid m=0)$ independent of $s$. Also $P(1,0 \mid 2,1) \le P(r=1 \mid s=2)$, and $P(1,0 \mid 2,0) \ge 0$, which is too loose.

A direct proof: consider the NS equality of Bob's marginal at $m=0$ and $m=1$. Work with the four probabilities $P(r,g \mid s=2,m)$ as a $2\times 2$ matrix for each $m$, rows summing to $P(r \mid 2)$. See solutions.md §7 if stuck after 20 minutes.

## 7.3 Saturating strategies (you must exhibit one)

The bound is useless for the paper unless it is tight: otherwise you have not "matched Silman *et al.*".

### Strategy A (device-independent, classical box)

Prepare Alice's box so that $r^c = s^c - 2$ deterministically (a classical box: output equals "the committed bit"). Guess $b = q$.

**Exercise 7.4.** Show that $q = r^c$ iff $ab = 0$, which has probability 3/4. When $q = r^c$, $q = b$. When $ab = 1$, $q = b \oplus 1$. Thus $P_{\text{gain}} = 3/4$.

### Strategy B (device-dependent, honest boxes)

Prepare the honest EPR boxes. Bob keeps box $\bar{c}$, inputs 0. Observe that honest settings $s^{\bar{c}} = 0$ and $s^c = 2$ are the *same* observable. Treat $q$ as Alice's output. If $r^{\bar{c}} = q$, guess she input 2; else guess 3.

**Exercise 7.5.** Show: if Alice input 2 ($b = 0$), Bob is always correct (ideal equality, and $q = r^c$). If Alice input 3, Bob's observable is orthogonal (in the $zx$-plane, $\pi/2$ from Alice's), so the guess is right with probability $1/2$. Average: $3/4$.

**Exercise 7.6.** Why include Strategy B in the paper at all, if A already saturates? Because it shows that even a Bob who follows the honest *state* preparation, and only cheats in the guess, already reaches the DI optimum. Security is not coming from "Bob cannot hold an EPR pair".

### 7.4 What this section does *not* use

- The value of $I_{\text{th}}$.
- The number of tests $N$.
- Quantum theory beyond NS (Strategy A is classical).

So Alice's security is the robust half of the paper. All the heavy machinery is Bob's security.

### Checkpoint

Prove $P_{\text{gain}} \leq 3/4$ on a whiteboard, then give Strategy A in three sentences. If you cannot expand (5) without the paper, redo Exercise 7.2.

## 8. Bob's security in the asymptotic limit

Paper analogue: §5.1–5.2. This is the technical core. Goal: a function $C(I)$ such that if the boxes at the commit/reveal step are known to satisfy $\mathbb{E}[I] \geq I_{\text{th}}$, then $P_{\text{cont}} \leq C(I_{\text{th}})$, with equality achieved by an explicit strategy.

Chapter 9 reduces the finite-$N$ memory case to this function.

### 8.1 Reduce Alice's instruments to one four-outcome POVM

Alice, at commit time, holds box $c$ and possibly an ancilla. A fully general strategy:

1. Commit: a two-outcome measurement producing $q$.
2. Reveal: a measurement that depends on $q$ and on the bit she now wishes to open, producing the announced $r^c$.

**Exercise 8.1.** Argue that if she wants to open 0, Bob requires $q = r^c$, so the reveal measurement for $b = 0$ is redundant: she might as well have set $r^c = q$ already. Therefore, in the reveal phase she only ever needs the two measurements associated with opening 1. Combine those with the commit measurement into a **single four-outcome POVM** $\{M^c_{kl}\}_{k,l\in\{0,1\}}$ on $\mathcal{H}^c$, with the interpretation:

- she sends $q = k$ at commit;
- if she opens 0, she announces $r^c = k$;
- if she opens 1, she announces $r^c = l$.

This is the paper's w.l.o.g. reduction. Write it carefully; referees look at this paragraph.

**Exercise 8.2.** Alice may know $c$ (an extra degree of freedom in the box can encode it). So $\mathcal{M}^0$ and $\mathcal{M}^1$ may differ. The paper notes that the optimum nevertheless uses the same strategy on both boxes, up to a local rotation of Bob's observables. A protocol that *always* sends the same box would *increase* $P_{\text{cont}}$ (Bob's random $c$ is essential). Record this as a remark, not as a lemma you must SDP.

### 8.2 Write $P_{\text{cont}}$ as a sum of four families of probabilities

Bob sends each box with probability 1/2.

**Opening** $b = 0$ ($s^c = 2$): token check forces $r^c = q = k$. Then Bob inputs $s^{\bar{c}} = 0$ and requires $r^{\bar{c}} = k$. Success probability

$$p_0 = \frac{1}{2}\sum_{k,l}\Big[P(r^1 = k,\,(k,l) \mid s^1 = 0, \mathcal{M}^0) + P(r^0 = k,\,(k,l) \mid s^0 = 0, \mathcal{M}^1)\Big].$$

**Opening** $b = 1$ ($s^c = 3$): token check allows any $r^c$ (i.e. $r^c = l$). Bob inputs $s^{\bar{c}} = 1$ and requires $r^{\bar{c}} = l$. Analogous $p_1$.

Then $P_{\text{cont}} = \frac{1}{2}(p_0 + p_1)$, which is the paper's (6).

**Exercise 8.3.** Derive (6) from the protocol abort conditions without looking. The factor 1/4 in (6) is $\frac{1}{2} \times \frac{1}{2}$: uniform $c$ and uniform intended $b$.

### 8.3 The optimization problem (7)

Maximize (6) over quantum realizations $\mathcal{Q} = \{\mathcal{H}^c, \rho, \{\Pi^c_{r|s}\}, \mathcal{M}^c\}_c$ subject to:

- CHSH of the two boxes $\geq I_{\text{th}}$,
- commutativity between operators on different boxes (tensor product, or commutation on a joint Hilbert space),
- POVM constraints: positivity and completeness for both $\Pi$ and $M$.

The paper's (7) writes this as a single trace:

$$P_{\text{cont}} = \frac{1}{4}\max_{\mathcal{Q}} \operatorname{Tr}\Big(\rho \sum_{c,k,l} M^c_{kl}(\Pi^{\bar{c}}_{k|0} + \Pi^{\bar{c}}_{l|1})\Big).$$

**Exercise 8.4.** Convince yourself that the operator sandwiched with $\rho$ is exactly the event "Bob's check passes", averaged over $c$ and over the two openings.

This problem is *not* a fixed-dimension SDP: $\dim \mathcal{H}$ is unknown. Relax it.

### 8.4 NPA hierarchy: what you actually compute

Apply NPA at **level 2** to (7). You obtain an upper bound $P^{\text{SDP2}}_{\text{cont}}(I_{\text{th}})$.

**Exercise 8.5.** List the operators that enter the level-2 moment matrix: products of up to two projectors among $\{\Pi^c_{r|s}, M^c_{kl}\}$. You do not need to code this to reconstruct the paper, but if you want Fig. 1 independently of the analytic strategy, use a package such as `ncpol2sdpa` (Python) or the original MATLAB+YALMIP+SeDuMi stack cited in the paper.

The paper's claim: the *analytic* strategy of §5.2 matches the level-2 SDP to $10^{-8}$, hence level 2 has already converged.

### 8.5 The saturating strategy (this you must derive by hand)

Guess the geometry before reading §5.2:

- Use one EPR pair $|\phi^+\rangle$.
- Bob's two check settings on a given box should be two axes in the $zx$-plane, **not necessarily** $\pi/2$ apart if $I < 2\sqrt{2}$.
- Alice, who received the other box, should measure **midway** between Bob's two axes, so that the angle to *either* check setting is the same. Then $p_0 = p_1$.
- One two-outcome measurement is enough (so the four-outcome POVM is overkill at the optimum). She announces that outcome as both $b$ and $r^c$ in the sense of the paper: she sends $b$ and $r^c$ equal to her measurement result.

**Exercise 8.6.** If two equatorial observables differ by an angle $\theta$, show that

$$P(\text{equal outcomes}) = \cos^2(\theta/2)$$

on $|\phi^+\rangle$. Therefore the strategy yields

$$P_{\text{cont}} = \cos^2(\theta/2).$$

At Tsirelson, Alice sits $\pi/4$ from each of two orthogonal axes, so $\theta = \pi/4$ and $P_{\text{cont}} = \cos^2(\pi/8)$. That is the GHZ number, obtained from CHSH geometry.

**Exercise 8.7.** Reproduce the paper's parameterization:

- Box 0, inputs 0 and 1: $\sigma_{2\theta}$ and $\sigma_z$.
- Box 1, inputs 0 and 1: $\sigma_{2\theta-\varphi}$ and $\sigma_{4\theta-\varphi}$.
- If Alice gets box 0, she measures $\sigma_{3\theta-\varphi}$; if box 1, she measures $\sigma_\theta$.

Draw Fig. 2: solid axes = Bob on box 0, dashed = Bob on box 1, dotted = Alice. All in the $zx$-plane. Check that Alice is always midway, with half-angle $\theta$.

**Exercise 8.8.** Using $\langle\sigma_\alpha \otimes \sigma_\beta\rangle = \cos(\alpha-\beta)$, compute the CHSH value of Bob's four observables (inputs 0, 1 on both boxes):

$$I = \langle\sigma_{2\theta} \otimes \sigma_{2\theta-\varphi} + \sigma_{2\theta} \otimes \sigma_{4\theta-\varphi} + \sigma_z \otimes \sigma_{2\theta-\varphi} - \sigma_z \otimes \sigma_{4\theta-\varphi}\rangle.$$

Simplify to the paper's (9):

$$I = 2\cos(2\theta-\varphi) - \cos(4\theta-\varphi) + \cos\varphi.$$

**Exercise 8.9.** For fixed $\theta$, maximize $I$ over $\varphi$. Show that a critical point is

$$\varphi_{\text{opt}} = \arccos\left(2\frac{\cos(2\theta)+\sin^2(2\theta)}{\sqrt{6-2\cos(4\theta)}}\right).$$

(Differentiate (9), solve $\partial I/\partial\varphi = 0$. The arccos form is a particular solution; check it against $\theta = \pi/4$, where you must get $\varphi_{\text{opt}} = \pi/4$ and $I = 2\sqrt{2}$.)

**Exercise 8.10.** Eliminate $\varphi$ to obtain a parametric curve $(I(\theta), P_{\mathrm{cont}}(\theta))$ for $\theta \in (0, \pi/2]$. This is Fig. 1. Implement it in `scripts/reconstruct_figures.py`.

Special values:

| $\theta$ | $P_{\mathrm{cont}}$ | $I$ (at $\varphi_{\mathrm{opt}}$) | meaning |
|---|---|---|---|
| $\pi/4$ | $\cos^2(\pi/8) \simeq 0.8536$ | $2\sqrt{2}$ | Tsirelson / GHZ match |
| $\to 0$ | $\to 1$ | $\to$ something $\leq 2$ | Alice nearly deterministic; CHSH cannot stay above 2 |

**Exercise 8.11.** Evaluate $I$ at $\theta \to 0$ with $\varphi_{\mathrm{opt}}$. You should see that to drive $P_{\mathrm{cont}} \to 1$ Alice must give up Bell violation, as expected.

### 8.6 How to write §5.1–5.2

Order in the paper (copy this structure):

1. W.l.o.g. four-outcome POVM.
2. Expression (6) for the two openings.
3. SDP (7) and NPA level 2.
4. Explicit strategy, (8)–(10), Fig. 2.
5. Statement that (8)–(10) saturate the SDP, hence Fig. 1 is tight.

Do not reverse 3 and 4: the SDP is the *upper* bound; the strategy is the *lower* bound; together they are the exact quantum value.

### Checkpoint

From memory: (i) why a four-outcome commit POVM is enough; (ii) $P_{\mathrm{cont}} = \cos^2(\theta/2)$; (iii) the measurement table of Exercise 8.7; (iv) why $\cos^2(\pi/8)$ appears.

## 9. Bob's security for finitely many tests

Paper analogue: §5.3 and Appendix D. Goal: if the devices have **arbitrary memory**, Alice's control is still bounded by something that tends to $C(I_{\mathrm{th}})$ as $N \to \infty$.

This is the Reichardt–Unger–Vazirani-style part of the paper, but much lighter: you only need a martingale on the CHSH indicators, not a full self-testing theorem.

### 9.1 Why i.i.d. is illegal

A dishonest Alice can program:

- a counter: "on use $N+1$, ignore CHSH and output a deterministic commitment";

- history-dependent measurements: fail CHSH on some paths, compensate on others;
- a strategy whose expected CHSH on round $n+1$ is *not* equal to the empirical $\bar{I}_n$.

So you cannot plug $\bar{I}_n$ into $C(\cdot)$ and call it a day. You need: empirical CHSH ≈ conditional expected CHSH on the *next* round, except with small probability, uniformly over strategies.

## 9.2 Express $P_{\text{cont}}$ as an average over $n$ and histories

Bob picks $n$ uniformly in $\{1,\dots,N\}$. For $n = N$, use $N+1$ is never CHSH-tested, so Alice can make $C = 1$ on that branch. Hence

$$P_{\text{cont}} = \frac{1}{N}\sum_{n=1}^{N-1}\sum_{\mathbf{w}_n} P(\mathbf{w}_n)\,\Theta(\bar{I}_n - I_{\text{th}})\,C\big(\mathbb{E}[I(W_{n+1})\mid \mathbf{w}_n]\big) + \frac{1}{N}\sum_{\mathbf{w}_N} P(\mathbf{w}_N)\,\Theta(\bar{I}_N - I_{\text{th}}).$$

The second sum is $\le 1/N$. This is (11).

**Exercise 9.1.** Derive (11) from the protocol and from the definition of $C(I)$. Identify the unit step $\Theta$ as “Bob did not abort in Phase 1”.

**Exercise 9.2.** Why is the $n = N$ term not $C(I_{\text{th}})/N$? Because Alice is not constrained at all on use $N+1$.

## 9.3 Lift every short history to a length-$(N-1)$ history

For each $\mathbf{w}_n$ with $n \le N-2$, consider compatible extensions $\mathbf{w}_{N-1}$. Rewrite the bound as an average over $\mathbf{w}_{N-1}$ of

$$\frac{1}{N}\sum_{n=1}^{N-1}\Theta(\bar{I}_n - I_{\text{th}})\,C\big(\mathbb{E}[I(W_{n+1})\mid \mathbf{w}_n]\big) + \frac{1}{N}.$$

This is (12).

**Exercise 9.3.** Write this change of summation carefully. The point is to have a *single* sample path on which you can mark the last time the empirical CHSH was still above threshold.

## 9.4 The last good time $K$

$$K(\mathbf{w}_{N-1}) = \max\{k \le N-1 : \bar{I}_k(\mathbf{w}_k) \ge I_{\text{th}}\}$$

(and, if you need a convention, $K = 0$ when the set is empty; those histories contribute 0 through $\Theta$).

Then $\Theta(\bar{I}_n - I_{\text{th}}) = 0$ for all $n > K$, so the inner sum over $n$ runs at most up to $K$. Histories that dipped below $I_{\text{th}}$ *before* $K$ only make the true $P_{\text{cont}}$ smaller, so dropping $\Theta$ for $n \le K$ upper-bounds:

$$P_{\text{cont}} \le \sum_k \sum_{\mathbf{w}:K(\mathbf{w})=k} P(\mathbf{w}) \frac{1}{N} \sum_{n=1}^{k} C(\mathbb{E}[I(W_{n+1}) \mid \mathbf{w}_n]) + \frac{1}{N}.$$

This is (14).

**Exercise 9.4.** Produce (14) from (12). The paper's inequality is because some $n < K$ may have $\Theta = 0$.

### 9.5 Split at $K_0 = \lceil (N-1)C(I_{\text{th}}) \rceil$ and use concavity

$C(I)$ is concave (Fig. 1 is concave; you may take this from the plot, or note that an SDP value as a function of a linear constraint is concave in the constraint). For $k \ge K_0$,

$$\frac{1}{k} \sum_{n=1}^{k} C(\mathbb{E}_n) \le C\Big(\frac{1}{k} \sum_{n=1}^{k} \mathbb{E}[I(W_{n+1}) \mid \mathbf{w}_n]\Big).$$

For $k < K_0$, bound $C \le 1$ and then $k/N \le (N-1)C(I_{\text{th}})/N$.

**Exercise 9.5.** This split looks like magic. Its only purpose is: the "small $k$" contribution cannot exceed about $C(I_{\text{th}})$ because there are few such terms, even if Alice sets $C = 1$ there. Check the arithmetic of the first sum in (15).

### 9.6 Typicality: empirical CHSH versus conditional expectations

Define the bad set $\pi_k(\varepsilon)$ of $k$-histories with

$$\bar{I}_k(\mathbf{w}_k) - \frac{1}{k} \sum_{n=1}^{k} \mathbb{E}[I(W_n) \mid \mathbf{w}_{n-1}] \ge \varepsilon.$$

On the complement, the argument of $C$ in the concave bound is at least $\bar{I}_k - \varepsilon \ge I_{\text{th}} - \varepsilon$ (because $K = k$ implies $\bar{I}_k \ge I_{\text{th}}$), hence $C(\cdots) \le C(I_{\text{th}} - \varepsilon)$ if $C$ is decreasing in a way... **wait**: $C(I)$ is *decreasing* in $I$ (more nonlocality, less control). So an *upper* bound on $C$ needs a *lower* bound on $I$. Yes: on good histories, average conditional CHSH $\ge \bar{I}_k - \varepsilon \ge I_{\text{th}} - \varepsilon$, so $C \le C(I_{\text{th}} - \varepsilon)$.

On bad histories, bound $C \le 1$.

**Exercise 9.6.** Assemble these into the first two displays of (19). Minimize over $\varepsilon \ge 0$. You should reach

$$P_{\text{cont}} \le \frac{N-1}{N} \min_{\varepsilon \ge 0} \Big[ C(I_{\text{th}} - \varepsilon) + (1 - C(I_{\text{th}} - \varepsilon)) Q(\varepsilon) \Big] + \frac{1}{N},$$

where $Q(\varepsilon)$ bounds $\sum_{k=K_0}^{N-1} P(\pi_k(\varepsilon))$.

## 9.7 Appendix D: Azuma–Hoeffding

**Exercise 9.7.** Complete Exercise 1.10: $Z_k = k\Delta_k$ is a martingale with bounded differences $D = 4 + 2\sqrt{2}$. Apply Azuma to get

$$P(\pi_k(\varepsilon)) \le \exp(-k\varepsilon^2/(2D^2)).$$

Sum a geometric series from $k = K_0$ to $N-1$:

$$Q(\varepsilon) = \frac{\exp(-K_0\varepsilon^2/(2D^2)) - \exp(-N\varepsilon^2/(2D^2))}{1 - \exp(-\varepsilon^2/(2D^2))}.$$

The paper also notes $D = (1-\cos^2(\pi/8))^{-1}$. **Exercise 9.8.** Check this numerical coincidence: $4 + 2\sqrt{2} \stackrel{?}{=} 1/\sin^2(\pi/8)$. (It is a curiosity, not used later.)

## 9.8 The limit $N \to \infty$

Choose $\varepsilon = \varepsilon(N)$ decaying *slower* than $N^{-1/2}$ (e.g. $N^{-1/3}$). Then $Q(\varepsilon) \to 0$, and the bound tends to $C(I_{\text{th}})$. For $I_{\text{th}} \to 2\sqrt{2}$, this is $\cos^2(\pi/8)$.

For Fig. 3 the paper uses (there is a **published inconsistency** between the figure caption and the main text)

- caption: $I_{\text{th}} = 2\sqrt{2}\,(1 - 1/\sqrt{N})$,
- text: $I_{\text{th}} = 2\sqrt{2} - 1/\sqrt{N}$.

**Exercise 9.9.** Plot both. They differ at finite $N$ but have the same limit. In your paper, pick one and use it in both the caption and the text. Numerically minimize (19) over $\varepsilon$ as in `scripts/reconstruct_figures.py`.

The Azuma tail is conservative ($D = 4 + 2\sqrt{2}$ is a crude diameter). Expect the bound (19) to sit well above $C(I_{\text{th}})$ until $N$ is huge; overlaying $C(I_{\text{th}})$ on Fig. 3 makes that gap visible. The published Fig. 3 is the numerical min of (19), not an experiment.

**Exercise 9.10.** The paper says the finite-$N$ bound is probably not tight because $Q(\varepsilon) \neq 0$. Do you agree? Where is the slack?

## 9.9 How to write §5.3 without drowning the reader

The published proof is one long chain of inequalities. When you reconstruct it, keep the *narrative* visible:

1. Average over $n$; isolate the untested last-round cheat $1/N$.
2. Look at a full path of length $N-1$; mark last time the test would have passed.
3. Concavity + split small/large $K$.
4. Martingale typicality: $\bar{I}$ cannot greatly exceed the mean of future-step expectations.
5. Azuma $\Rightarrow Q(\varepsilon) \to 0$.

If a reader remembers only those five bullets, they can rebuild (11)–(19).

### Checkpoint

Write the five-bullet narrative. Reproduce $D = 4 + 2\sqrt{2}$ and the form of (19) without looking. Run `python scripts/reconstruct_figures.py` and confirm Fig. 3 approaches $\cos^2(\pi/8)$.

# 10. Appendices A–C (and what to do with D)

Appendix D was already the lemma for chapter 9. Here you reconstruct the three *protocol* variants.

## 10.1 Appendix A — PR-box bit commitment

### Why this appendix exists

Two opposing intuitions:

- Dishonest parties get stronger correlations ⇒ worse security.
- Honest parties get pseudo-telepathy back ⇒ GHZ-style "test and check with the same measurement", no sequential CHSH, possibly better security.

You will find $P_{\text{gain}} \le 3/4$ (unchanged, still NS) and $P_{\text{cont}} \le 3/4$ (improved, and now balanced).

Do **not** assume the PR box is trusted hardware (that would be Buhrman *et al.* 2006, which gets *perfect* BC if dishonest parties cannot tamper). Here Alice prepares any no-signalling box.

### Protocol (write it yourself)

Alice holds box 0, Bob box 1. Two inputs, two outputs. Honest PR correlation $r^0 \oplus r^1 = s^0 s^1$.

1. **Commit.** Alice inputs $s^0 = b$, samples pad $a$, sends $q = r^0 \oplus a s^0$.
2. **Reveal.** Alice sends $s^0, r^0$. Bob checks the token ($q = r^0$ or $q = r^0 \oplus s^0$). Then he picks uniform $s^1$ and checks the PR equation. Abort if either fails.

**Exercise 10.1.** Why is sequential testing gone? Because the PR equation is *certain*. One random $s^1$ both tests the box and tests consistency of the revealed $(s^0, r^0)$. That is the GHZ trick, restored.

### Alice's security

**Exercise 10.2.** Repeat §4 with inputs $\{0, 1\}$ instead of $\{2, 3\}$. Same NS bound $P_{\text{gain}} \le 3/4$.

**Exercise 10.3.** Saturating strategy: assume $q = r^0$, input $s^1 = 1$, guess $g = r^0 \oplus r^1$. Check the three cases $(s^0, a) = (0, *), (1, 0), (1, 1)$. You should get exactly $3/4$.

### Bob's security

Alice can prepare any NS box, possibly after deciding $q$ (she sends $q$ without receiving anything). Relabelling $q \to q \oplus 1$, $r^0 \to r^0 \oplus 1$ is a symmetry, so set $q = 0$.

**Exercise 10.4.** If she opens 0, she must send $r^0 = 0$. Bob then checks $r^1 = 0$ for random $s^1$. Success probability $\frac{1}{2}[P(r^1 = 0 \mid s^1 = 0) + P(r^1 = 0 \mid s^1 = 1)]$.

If she opens 1, both values of $r^0$ are allowed; Bob checks the PR equation. Write that probability, add the two openings with weight 1/2, and obtain the paper's (21).

**Exercise 10.5.** Using only NS and positivity, prove (21) $\leq 3/4$. Then give the saturating strategy: Bob's box is classical, $P(r^1 = 0 \mid s^1) = 1$ for both $s^1$.

### How to discuss this in §6 / App. A

The "supra-quantum structure allows for greater security" claim is: $P_{\text{cont}}$ dropped from 0.85 to 0.75, the protocol is balanced, and one no longer needs $N \to \infty$. $P_{\text{gain}}$ did not get worse.

---

## 10.2 Appendix B — free reveal time, worse $P_{\text{cont}}$

### The modification

Bob still CHSH-tests $n$ rounds, still sends box $c$. **But** at time $t_{n+1}$ he *already* inputs a random bit $d$ into the kept box, *before* Alice has revealed (in fact, he can do this even before she has committed, as long as it is at $t_{n+1}$). Alice may reveal at any later time. Bob's equality check is performed **only if** $d$ happens to equal the revealed $b$.

**Exercise 10.6.** Write the three phases with coins $c, d$. Check that $t_{i+1} - t_i$ no longer needs to contain Alice's round-trip: Bob measures the kept box on a clock, independently of Alice. Intervals may be arbitrarily short.

### Security

**Exercise 10.7.** Alice's security is unchanged ($P_{\text{gain}}$).

**Exercise 10.8.** For Bob's security: with probability 1/2, $d \neq b$ and Bob skips the last test, so he accepts any opening that merely satisfies the *classical* token check. A dishonest Alice can then open both bits (the token $q = r \oplus ab$ is designed to be openable both ways). With probability 1/2, $d = b$ and the situation is the original protocol. Conclude

$$P_{\text{cont}} \longmapsto \frac{1}{2}(P_{\text{cont}} + 1).$$

Asymptotically this is $\frac{1}{2}(\cos^2(\pi/8) + 1) \simeq 0.927$. State this number in your notes; the paper leaves it in functional form.

---

## 10.3 Appendix C — large office: free reveal time, same $P_{\text{cont}}$

### Resources

$N+1$ *pairs* of boxes, pairwise isolated from each other (not only the two boxes of a pair, but different pairs as well). Impractical; it is an existence argument.

### Protocol (write it yourself)

1. Bob picks private $n \in \{1, \dots, N+1\}$ and coin $c$, sends box $c$ of pair $n$.
2. Alice commits as usual on that box.
3. Alice reveals at a time of her choosing.
4. Bob, *in parallel*, CHSH-tests **all other pairs** and, simultaneously, measures the kept box of pair $n$ with input $b$. Abort on equality failure or if the CHSH average over the $N$ tested pairs is below $I_{\text{th}}$.

**Exercise 10.9.** Why does free reveal time not let the kept box "know" it is special? Because the CHSH tests are on *other* pairs, and the check measurement on pair $n$ is simultaneous with those tests—no extra delayed measurement on a box that already saw $n$ CHSH rounds.

### Reduction to the sequential bound

You need not re-prove Azuma. Argue by two relaxations, each of which can only *help* Alice:

1. Estimate CHSH using only pairs $1, \dots, n-1$ (Alice knows the numbering). Worse for Bob, so $P_{\text{cont}}$ does not decrease.
2. Give every box in pair $k$ full memory of all previous same-side boxes. This recreates a sequential memory process. Worse for Bob again.

Then $P_{\text{cont}}$ is at most that of the original sequential protocol. Timing issues disappear because tests are simultaneous.

**Exercise 10.10.** Write this comparison argument as three paragraphs. It is the entire App. C security proof.

---

## 10.4 What *not* to put in the appendices

- Do not redo §4 in App. B/C beyond one sentence ("unchanged").
- Do not derive a new finite-$N$ rate for App. C; the paper explicitly refuses to.
- Do not claim App. B intervals can be short *and* $P_{\text{cont}}$ stays $\cos^2(\pi/8)$. That is App. C's job.

## Checkpoint

State the three appendix trade-offs in a table:

| Variant | Reveal time | Extra resources | $P_{\text{gain}}$ | $P_{\text{cont}}$ |
|---|---|---|---|---|
| Main protocol | fixed | 1 pair, sequential | 3/4 | $C(I_{\text{th}}) + o(1)$ |
| App. A | free (single shot) | PR box | 3/4 | 3/4 |
| App. B | free | 1 pair | 3/4 | $\frac{1}{2}(C+1)$ |
| App. C | free | $N+1$ pairs in parallel | 3/4 | ≤ sequential bound |

# 11. Write the paper

Use this chapter only after the reconstruction checklist is done. The goal is to produce a manuscript with the same *claims and structure* as Aharon *et al.*, NJP **18**, 025014 (2016), in your own words.

## 11.1 Title, authors, keywords

- Title can stay descriptive: device-independent, bit commitment, CHSH.
- Keywords used by NJP: quantum cryptography, device-independent quantum information, bit commitment, nonlocality.

## 11.2 Abstract (one paragraph)

Must contain:

1. Previous DI BC/CF used GHZ; unique among bipartite DI tasks in that respect.
2. GHZ pseudo-telepathy was essential to those proofs.
3. This work: CHSH testing, same cheating probabilities as Silman *et al.*, devices with memory.
4. Caveat: fixed reveal time.
5. Post-quantum recasting: overall more security.

Target length: ~150–180 words. Do not put formulae other than perhaps the two numbers 0.8536 and 0.75 (the published abstract is formula-free; the introduction has the numbers).

## 11.3 Section 1 — Introduction

Paragraph plan (the published order is good; steal the *order*, not the sentences):

1. Cryptographic assumptions in general; quantum vs classical; device-dependent vs DI.
2. DI via Bell tests; DIQKD as the flagship; hacking motivation.
3. Other DI tasks (randomness, self-testing, estimation, multipartite entanglement).
4. Distrustful cryptography as the extra challenge (conflicting goals).

5. Silman11 / Aharon14: GHZ; open question whether the whole distrustful class is DI.
6. Why GHZ was used (pseudo-telepathy; same measurements for test and check).
7. Why CHSH is the natural next question (RUV; experimental EPR vs GHZ).
8. **This work:** protocol, numbers, memory, imperfect devices, fixed reveal time, coin-flipping still OK, Apps. B–C, App. A.
9. Roadmap of sections.

Citations you need (minimum): Mayers–Yao; Barrett–Hardy–Kent; Clauser *et al.*; Acín *et al.* DIQKD; Pironio *et al.* 2009; Reichardt–Unger–Vazirani; Lo–Chau; Mayers; Spekkens–Rudolph; Chailloux–Kerenidis; Silman *et al.* 2011; Greenberger–Horne–Zeilinger; Gisin–Méthot–Scarani (no bipartite pseudo-telepathy); Kent relativistic BC.

## 11.4 Section 2 — Background

**2.1 Bit commitment.** Definitions of phases, $P_{\text{cont}}$, $P_{\text{gain}}$, perfect, balanced, CK bound.

**2.2 Device-independence.** Five assumptions; formula (1); memory/clocks; meaning of "send a box"; shielding vs relativity.

## 11.5 Section 3 — Protocol

- Notation paragraph ($S^i_k$, $W_k$, $\sigma_\theta$).
- Honest correlations (2) and the equality pairs; measurement table.
- Noisy case.
- Numbered protocol 1–3 with times $t^a, t^b, t^c, t^d, t_i$.
- Formulae (3)–(4) for $\bar{I}_n$ and $I(W_k)$.
- Completeness discussion (statistical abort vs noisy reveal abort).
- Timing argument (the kept box must not know it is in reveal).
- "Not strictly BC" paragraph + pointer to Apps. B–C.

Footnotes to restore: private *n*; factor 4 in the indicator; interval $(t_i, t_{i+1}]$.

## 11.6 Section 4 — Alice's security

- General Bob strategy; display (5); NS ⇒ 3/4.
- Two saturating strategies (classical box; honest EPR + input 0).
- One sentence: matches Silman *et al.*

## 11.7 Section 5 — Bob's security

Tell the reader the three-subsection plan in the first paragraph.

**5.1** Four-outcome POVM; (6); SDP (7); NPA level 2; Fig. 1 (produced from 5.2).

**5.2** Explicit strategy; (8)–(10); Fig. 2; saturation of the SDP.

**5.3** Memory; (11)–(19); Fig. 3; limit $\cos^2(\pi/8)$. Move the martingale lemma to App. D.

### 11.8 Section 6 — Summary

Restate: pseudo-telepathy is not essential; sequential CHSH + hidden $n$ + fixed times suffice; memory included; experimental motivation (EPR vs GHZ); techniques should transfer to DI coin flipping without BC and to DI OT.

### 11.9 Figures

| Figure | Content | How to produce |
|---|---|---|
| 1 | $P_{\text{cont}}$ vs $I_{\text{th}}$ | parametric plot of (8)–(10) |
| 2 | axes in the $zx$-plane | schematic (solid / dashed / dotted) |
| 3 | finite-$N$ bound vs $\log_{10} N$ | numerical min of (19) |

Run `python scripts/reconstruct_figures.py`. The published Fig. 1 is the analytic curve; Fig. 3 is a numerical upper bound, not an experiment.

### 11.10 Tone and claims to get right

- Do not claim perfect BC.
- Do not claim the protocol is balanced.
- Do not claim Alice-chosen reveal time in the main protocol.
- Do not claim the finite-$N$ bound is tight.
- Do claim matching Silman *et al.* **in the infinite-test limit**.
- Do claim shielding instead of spacelike separation.
- Do mention that always sending the same box would help Alice.

### 11.11 After the draft exists

1. Check every numbered equation against reconstruction-checklist.md.
2. Check that Fig. 1 saturates $\cos^2(\pi/8)$ at $I = 2\sqrt{2}$.
3. Check that Fig. 3's $I_{\text{th}}(N)$ matches its caption.
4. Verify you did not mix arXiv and NJP wording on the relativistic paragraph; pick one version.
5. Acknowledgements if you used YALMIP/SeDuMi or a Python SDP solver.

## Equation map (paper → meaning)

Use this as an index while writing. Every display in NJP **18**, 025014 (2016) should appear in your manuscript, in this logical order.

| Eq. | Location | What it is | You derive it in |
|---|---|---|---|
| (1) | §2.2 | Born rule for two isolated boxes | ch. 4 |

| Eq. | Location | What it is | You derive it in |
|---|---|---|---|
| (2) | §3 | Honest CHSH $= 2\sqrt{2}$ on inputs $\{0, 1\}$ | ch. 5 |
| (3) | §3 | Empirical CHSH $\bar{I}_n$ | ch. 6 |
| (4) | §3 | Single-round CHSH indicator (prefactor 4) | ch. 1, 6 |
| (5) | §4.1 | $P_{\text{gain}}$ expanded, then $\leq 3/4$ | ch. 7 |
| (6) | §5.1 | $P_{\text{cont}}$ as four families of probabilities | ch. 8 |
| (7) | §5.1 | SDP/NPA optimization of (6) at fixed $I_{\text{th}}$ | ch. 8 |
| (8) | §5.2 | $P_{\text{cont}} = \cos^2(\theta/2)$ | ch. 8 |
| (9) | §5.2 | $I(\theta, \varphi)$ for the cheating axes | ch. 8 |
| (10) | §5.2 | $\varphi_{\text{opt}}(\theta)$ | ch. 8 |
| (11) | §5.3 | $P_{\text{cont}}$ averaged over $n$, plus $1/N$ | ch. 9 |
| (12) | §5.3 | Same, written over length-$(N-1)$ histories | ch. 9 |
| (13) | §5.3 | Last good time $K(\mathbf{w}_{N-1})$ | ch. 9 |
| (14) | §5.3 | Bound using $n \leq K$ only | ch. 9 |
| (15) | §5.3 | Split at $K_0$, concavity of $C$ | ch. 9 |
| (16) | §5.3 | Bad set $\pi_k(\varepsilon)$ | ch. 9 |
| (17) | App. D | Azuma tail on $\pi_k(\varepsilon)$ | ch. 9 |
| (18) | §5.3 | Sum of tails $Q(\varepsilon)$ | ch. 9 |
| (19) | §5.3 | Final finite-$N$ bound | ch. 9 |
| (20) | App. A | PR-box relation $r^0 \oplus r^1 = s^0 s^1$ | ch. 10 |
| (21) | App. A | Alice's control polynomial for PR boxes | ch. 10 |
| (22) | App. D | Martingale increment $\Delta_k$ | ch. 9 |
| (23) | App. D | Bounded differences $\lvert Z_{k+1} - Z_k \rvert \leq D$ | ch. 9 |

| Eq. | Location | What it is | You derive it in |
|---|---|---|---|
| (24) | App. D | Azuma–Hoeffding statement | ch. 9 |

Figures:

| Fig. | Content | Script |
|---|---|---|
| 1 | $C(I_{\text{th}})$ from (8)–(10) | `scripts/reconstruct_figures.py` |
| 2 | Cheating measurement axes | draw by hand from ch. 8 |
| 3 | Finite-$N$ bound vs $\log_{10} N$ | `scripts/reconstruct_figures.py` |

## Reconstruction checklist

Tick only what you have derived yourself. When everything is ticked, go to 11-write-the-paper.md.

### Definitions

- ☐ $P_{\text{cont}} = \frac{1}{2}(p_0 + p_1)$ and $P_{\text{gain}}$ written in words and symbols
- ☐ Five DI assumptions listed
- ☐ Formula (1) for product measurements
- ☐ "Sending a box" defined operationally
- ☐ Shielding vs relativistic BC: three differences

### Honest physics

- ☐ Measurement table (box 0 / box 1, inputs 0–3)
- ☐ CHSH value $2\sqrt{2}$ verified with $\langle \sigma_\alpha \otimes \sigma_\beta \rangle = \cos(\alpha - \beta)$
- ☐ Equality pairs $(i, i + 2 \bmod 4)$ verified
- ☐ Commit/reveal rule $s^c = b + 2$, $s^{\bar{c}} = b$ matches those pairs for all $c, b$

### Protocol

- ☐ Private uniform $n \in \{1, \ldots, N\}$ and why it must be private
- ☐ Indicator $I(W_k)$ with prefactor 4, and $\bar{I}_n$
- ☐ Abort if $\bar{I}_n < I_{\text{th}}$
- ☐ Coin $c$, send box $c$ at $t^b < t_{n+1}$
- ☐ $q = r^c \oplus ab$ at $t^c$
- ☐ Reveal $(b, r^c)$ before $t_{n+1}$
- ☐ Token check: $q = r$ or $q = r \oplus b$
- ☐ Check measurement at $t_{n+1}$ with input $b$ on the kept box
- ☐ Timing constraint: interval long enough for the round trip
- ☐ Honest completeness (ideal and noisy) discussed

## Alice's security (§4)

- ☐ Display (5) expanded from the protocol
- ☐ NS inequalities written
- ☐ $P_{\text{gain}} \leq 3/4$
- ☐ Strategy A (deterministic box, guess $q$)
- ☐ Strategy B (honest EPR, Bob inputs 0)

## Bob's security, asymptotic (§5.1–5.2)

- ☐ Four-outcome POVM reduction
- ☐ Formula (6) for $p_0, p_1$
- ☐ Optimization (7) including CHSH constraint and commutators
- ☐ NPA level-2 mentioned as an upper bound
- ☐ Strategy: angles $2\theta$, $z$, $2\theta - \varphi$, $4\theta - \varphi$; Alice $3\theta - \varphi$ / $\theta$
- ☐ $P_{\text{cont}} = \cos^2(\theta/2)$
- ☐ Formula (9) for $I(\theta, \varphi)$
- ☐ $\varphi_{\text{opt}}$ as in (10); check $\theta = \pi/4 \Rightarrow I = 2\sqrt{2}$
- ☐ Fig. 1 reproduced
- ☐ Statement that the strategy saturates the SDP

## Bob's security, finite $N$ (§5.3, App. D)

- ☐ Formula (11) with the $1/N$ last-round term
- ☐ Definition of $K(\mathbf{w}_{N-1})$
- ☐ Concavity split at $K_0 = \lceil (N-1)C(I_{\text{th}}) \rceil$
- ☐ Bad set $\pi_k(\varepsilon)$
- ☐ Martingale $Z_k$ and $D = 4 + 2\sqrt{2}$
- ☐ Azuma bound (17) and sum $Q(\varepsilon)$
- ☐ Final bound (19)
- ☐ Limit $N \to \infty$ recovers $C(I_{\text{th}})$
- ☐ Fig. 3 reproduced; caption consistent with the choice of $I_{\text{th}}(N)$

## Appendices

- ☐ App. A protocol, both bounds = 3/4, classical saturating strategy for Alice
- ☐ App. B: random $d$, $P_{\text{cont}} \to \frac{1}{2}(P_{\text{cont}} + 1)$, short intervals allowed
- ☐ App. C: $N + 1$ pairs, reduction to sequential bound in two relaxations
- ☐ Trade-off table (fixed time vs control vs number of boxes vs PR boxes)

## Writing

- ☐ Abstract contains GHZ history, CHSH result, memory, numbers or equivalent, reveal-time caveat, post-quantum remark
- ☐ No claim of perfect or balanced quantum BC in the main protocol
- ☐ All numbered equations (1)–(24) have a counterpart in your notes

## Unicode and LaTeX math

The study guide uses GitHub-style math (`$...$` inline, `$$...$$` display). That renders on GitHub, in Cursor's markdown preview, and in the built HTML/PDF.

### View rendered math

| How | Command / place |
|---|---|
| GitHub / Cursor preview | open any `study-guide/*.md` file |
| HTML (KaTeX) | `bash scripts/build_study_guide.sh` then open `study-guide/build/study-guide.html` |
| PDF (XeLaTeX + Unicode math font) | same script; open `study-guide/build/study-guide.pdf` |

XeLaTeX is installed with **TeX Live**, **fontspec**, and **unicode-math**. Body text uses DejaVu; formulae use **TeX Gyre DejaVu Math**, so Greek letters, operators, and subscripts are real Unicode glyphs in the PDF.

### Symbol card (visible even in raw text)

Use these when you write notes by hand or in a plain editor.

| Meaning | Unicode | LaTeX |
|---|---|---|
| pi | π | `\pi` |
| theta | θ | `\theta` |
| phi | φ | `\varphi` |
| sigma | σ | `\sigma` |
| rho | ρ | `\rho` |
| psi | ψ | `\psi` |
| alpha | α | `\alpha` |
| epsilon | ε | `\varepsilon` |
| less-or-equal | ≤ | `\le` |
| greater-or-equal | ≥ | `\ge` |
| not-equal | ≠ | `\ne` |
| much-less | ≪ | `\ll` |
| approximately | ≈ | `\approx` |
| isomorphic / simeq | ≃ | `\simeq` |
| element of | ∈ | `\in` |
| subset | ⊂ | `\subset` |
| tensor | ⊗ | `\otimes` |
| XOR / plus-mod-2 | ⊕ | `\oplus` |
| times | × | `\times` |
| cdot | · | `\cdot` |
| plus-minus | ± | `\pm` |

| Meaning | Unicode | LaTeX |
|---|---|---|
| infinity | ∞ | `\infty` |
| square root | √ | `\sqrt` |
| sum | ∑ | `\sum` |
| product | ∏ | `\prod` |
| integral | ∫ | `\int` |
| right arrow | → | `\to` |
| implies | ⇒ | `\Rightarrow` |
| bra / ket | ⟨ ⟩ | `\langle \rangle` |
| real numbers | ℝ | `\mathbb{R}` |
| complex numbers | ℂ | `\mathbb{C}` |
| identity | 𝟙 | `\mathbb{1}` |
| empty set | ∅ | `\emptyset` |
| for all | ∀ | `\forall` |
| exists | ∃ | `\exists` |
| hbar | ℏ | `\hbar` |
| dagger | † | `\dagger` |
| circ / composition | ∘ | `\circ` |
| similar | ∼ | `\sim` |
| proportional | ∝ | `\propto` |

Paper-specific shortcuts:

| Meaning | Unicode-ish | LaTeX |
|---|---|---|
| Alice’s control | P_cont | `$P_{\mathrm{cont}}$` |
| Bob’s information gain | P_gain | `$P_{\mathrm{gain}}$` |
| CHSH value | I | `$I$` |
| Tsirelson bound | 2√2 | `$2\sqrt{2}$` |
| GHZ / midway angle | cos²(π/8) | `$\cos^2(\pi/8)$` |

### Rebuild after editing chapters

```
python3 scripts/convert_math_delimiters.py study-guide/*.md   # if you typed \( ... \)
bash scripts/build_study_guide.sh
```